Confidence Intervals & Hypothesis Testing

ANSWER KEY

Section 1: Constructing & Interpreting Confidence Intervals

1 What is the correct interpretation of a 95% confidence interval for a population mean $(42.1, 48.7)$?

a. **We are 95% confident that the true population mean falls between 42.1 and 48.7** ✓

b. There is a 95% probability that the true mean is between 42.1 and 48.7

c. 95% of all individual values in the population lie between 42.1 and 48.7

d. 95% of sample means will equal the population mean

Confidence level refers to the long-run success rate of the method: in repeated sampling, 95% of intervals generated will capture the fixed true parameter.

2 In a sample of 100 items, $\hat{p} = 0.60$. With critical value $z^* = 1.96$, what is the margin of error for a 95% confidence interval for p?

a. $1.96\sqrt{\frac{0.60(0.40)}{100}} \approx 0.096$ ✓

b. $1.96\sqrt{\frac{0.50(0.50)}{100}} \approx 0.098$

c. 0.60 ± 0.05

d. 0.049

$$\text{ME} = z^* \times \text{SE} = 1.96\sqrt{\frac{0.60(0.40)}{100}} = 1.96\sqrt{0.0024} = 1.96(0.04899) \approx 0.096.$$

3 How can you decrease the width of a confidence interval?

a. **Increase the sample size or decrease the confidence level** ✓

b. Increase the confidence level or increase the sample size

c. Decrease the sample size or increase the confidence level

d. Increase the standard deviation of the data

Width is $2 \times \text{critical value} \times \frac{\sigma}{\sqrt{n}}$. Increasing n shrinks the denominator; lowering confidence level (e.g. from 99% to 90%) lowers the critical value.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Section 2: Hypothesis Testing Logic & P-values

4 What is the statistical definition of a P-value?

a. **The probability of observing a test statistic as extreme or more extreme than the one observed, assuming the null hypothesis is true ✓**

b. The probability that the null hypothesis is true

c. The probability that the alternative hypothesis is false

d. The probability of making a Type I error

The P-value is calculated under the assumption that H_0 is true, measuring how plausible the observed sample outcome is purely by chance.

5 If a test yields a P-value of 0.023 and the significance level is $\alpha = 0.05$, what is the conclusion?

a. **Reject H_0; there is convincing statistical evidence in favor of H_a ✓**

b. Fail to reject H_0; the evidence is insufficient

c. Accept H_0 as definitively proven

d. Reject H_a; the test is inconclusive

Because $\text{P-value} = 0.023 < \alpha = 0.05$, we reject H_0 and conclude that there is statistically significant evidence for H_a.

6 Why do statisticians state 'fail to reject H_0' instead of 'accept H_0'?

a. **Failing to find sufficient evidence against H_0 does not prove that H_0 is true ✓**

b. Because H_0 is always false in practice

c. To avoid computing the test statistic

d. Because the P-value is always approximate

Absence of evidence is not evidence of absence; we never prove the null hypothesis true, we merely conclude the data is consistent with it.

Section 3: Type I Error, Type II Error, and Power

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 What is a Type I error?

a. **Rejecting the null hypothesis when the null hypothesis is actually true ✓**

b. Failing to reject the null hypothesis when the null hypothesis is actually false

c. Calculating an incorrect test statistic

d. Using the wrong degrees of freedom

A Type I error is a false positive: rejecting a true null hypothesis. The probability of a Type I error equals the significance level α.

8 What is a Type II error?

a. **Failing to reject the null hypothesis when the null hypothesis is actually false ✓**

b. Rejecting the null hypothesis when it is true

c. Selecting a biased sample

d. Misinterpreting the P-value

A Type II error (probability β) is a false negative: failing to detect a real effect when the alternative hypothesis is true.

9 What is the power of a statistical test?

a. **The probability of correctly rejecting a false null hypothesis ($1 - \beta$) ✓**

b. The probability of making a Type I error (α)

c. The sample size n

d. The probability of retaining a true null hypothesis

Power $= 1 - \beta$ represents the test's ability to successfully detect a true difference or effect of a specified magnitude.

Section 4: One-Sample and Two-Sample t-Procedures

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 When estimating a population mean μ with unknown σ, which distribution and test statistic are used?

a. **t-distribution with $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ and $df = n - 1$** ✓

b. Standard normal distribution with $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

c. Chi-square distribution with $df = n - 1$

d. F-distribution with $df = (n - 1, n - 2)$

When σ is replaced by the sample standard deviation s, the standardized statistic follows Student's t -distribution with $n - 1$ degrees of freedom.

11 How does the t-distribution compare to the standard normal (Z) distribution?

a. **The t-distribution is symmetric and bell-shaped but has heavier tails (more spread)** ✓

b. The t-distribution is skewed right

c. The t-distribution has less variability than Z

d. The t-distribution is bounded between -3 and 3

The t-distribution has heavier tails to account for the extra uncertainty introduced by estimating σ with s. As $df \to \infty$, t approaches Z.

12 A researcher compares scores before and after a training program on the same 20 subjects. Which test should be performed?

a. **Paired t-test on the differences ($d_i = \text{After}_i - \text{Before}_i$)** ✓

b. Two-sample independent t-test

c. Two-sample z-test for proportions

d. Chi-square test of independence

Because the two observations come from the same individuals, the data is dependent (matched pairs), requiring a one-sample paired t-test on the pairwise differences.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647